SOFT3111 Internet and Web Programming

Clinic Appointment Assistant

Web Application Project Proposal

Project 1

Group 2

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GitHub repository URL

https://github.com/fatihemreakardere/SOFT3111-Group02-ClinicAppointmentAssistant

1 Project Overview and Motivation

Clinic Appointment Assistant is a web application that will help patients arrange clinic visits through a conversational chatbot. The proposed clinic has five departments: Ear, Nose and Throat (ENT), Internal Medicine, Dermatology, Pulmonology, and Psychiatry. Each department has three doctors, giving a total of 15. Patients will use personal accounts to book, change, or cancel appointments, while doctors and clinical leaders will use dedicated views to follow schedules and workloads.

The project addresses the difficulty of finding a suitable appointment when patients must compare several doctors and available times. A requested doctor may be fully booked, a preferred time may fall outside clinic hours, or another available doctor may be overlooked. The chatbot will guide patients through these situations, explain why a request cannot be accepted, and suggest valid alternatives. It will also provide the clinic telephone number, address, and a maps link, making practical information available alongside booking assistance.

Patients will be able to request either a particular doctor or a department. For department requests, the system will select an available doctor with the fewest appointments on the requested day. All appointments will have the same duration, and tied workloads will be resolved by rotating assignments. This approach will distribute bookings more evenly while preserving a patient's choice when a specific doctor is requested. Patients will be able to access the service at any time, but appointments will only take place during clinic opening hours and doctor availability.

Confirmed appointments will appear in the relevant doctor's schedule, and the planned Google Calendar integration will provide patients with a calendar event link. Doctors will be able to review their commitments, Department Heads will oversee their own departments, and a Chief Medical Officer will review clinic-wide monthly activity. The expected benefit is a clearer booking process for patients and better visibility of appointment demand for staff. The semester project will focus on scheduling, account access, and operational reporting; diagnosis, prescriptions, payments, and electronic medical records are outside its scope.

2 Target Users and User Roles

Each department has one Department Head chosen from its three doctors. One of the 15 doctors also serves as Chief Medical Officer. These responsibilities do not increase the doctor count.

Role	Expectations and permitted actions
Patient	Find a suitable visit; manage only their own appointments through chat and access clinic information and calendar links.
Doctor	See an accurate personal calendar, upcoming appointments, free slots, appointment counts, and booked hours.
Department Head	Retain doctor access and view schedules and workloads for the three doctors in their own department.
Chief Medical Officer	Retain personal doctor access and view clinic-wide monthly changes, doctor workloads, and total scheduled hours.

3 Functional Requirements

High priorities cover the booking workflow and doctor access. Medium priorities add departmental oversight, external calendar links, reporting, and clinic information.

ID	Required system behavior	Priority
FR-01	Patients shall create an account, log in, and access only their own appointments. Staff shall log in with their assigned roles.	High
FR-02	The chatbot shall collect a department or doctor, date, and preferred time, asking for clarification when required details are missing or ambiguous.	High
FR-03	The system shall use one fixed appointment duration, reject occupied or out-of-hours slots, explain the conflict, and suggest available alternatives. Booking requests may be made at any time.	High
FR-04	For department bookings, the system shall select a doctor available for the chosen slot with the fewest non-cancelled appointments that day, rotating assignments when tied. Specific-doctor requests shall preserve that choice.	High
FR-05	Before booking, the chatbot shall show the doctor, department, date, and time for patient confirmation. It shall recheck availability, save one appointment, and display it in the assigned doctor’s schedule.	High
FR-06	Patients shall cancel their own future appointments after confirmation. The system shall mark them cancelled, release the slot, and update the doctor’s schedule.	High
FR-07	Patients shall reschedule their own future appointments to a valid free slot after confirmation. The original booking shall remain in place if the replacement cannot be saved.	High
FR-08	Doctors shall view their personal daily and weekly calendars, including booked and available slots, appointment counts, and total booked hours for the selected period.	High
FR-09	Department Heads shall view schedules and workload summaries for doctors in their own department, including appointment counts and booked hours.	Medium
FR-10	The system shall use the Google Calendar API to create an event for a confirmed booking and give its link to the patient. Rescheduling and cancellation shall update the corresponding event.	Medium
FR-11	The Chief Medical Officer shall select a month and view clinic appointment totals, percentage change from the previous month, individual doctor workloads, and total scheduled appointment hours.	Medium
FR-12	The chatbot shall provide the configured clinic telephone number, address, maps link, and opening hours when requested.	Medium

4 Non Functional Requirements

The following are proposed acceptance targets for the semester application.

ID	Quality	Verifiable acceptance target
NFR-01	Security	Protected requests shall enforce account ownership and staff permissions on the server. Tests shall deny access to another patient's bookings and another department's restricted data. Passwords shall be hashed and connections encrypted with HTTPS.
NFR-02	Reliability	Two simultaneous requests for the same doctor and slot shall produce at most one confirmed appointment. Saved appointments shall survive an application restart. A failed rescheduling operation shall preserve the original appointment.
NFR-03	Performance	With 30 concurrent test users and the 15-doctor dataset, at least 95% of availability searches shall finish within 2 seconds and calendar views within 3 seconds, excluding external service latency.
NFR-04	Usability	Booking, cancellation, and rescheduling shall show a clear confirmation and result. At 360-pixel mobile width and 1280-pixel desktop width, core screens shall remain readable without horizontal page scrolling.
NFR-05	Accessibility	Core workflows shall support keyboard-only use, visible keyboard focus, labelled inputs, and errors expressed in text. Normal text shall have a contrast ratio of at least 4.5 to 1.
NFR-06	Integration recovery	If Google Calendar is unavailable, the clinic booking shall remain recorded and its calendar-sync state shall be shown as pending. Retries shall synchronize the appointment without creating duplicate events.

Scheduling and allocation rules

One clinic-wide appointment duration will be configured before use. A valid slot must fit entirely inside clinic opening hours and the doctor's available schedule. All dates and times will use the clinic's configured time zone. After finding availability, the system will count each eligible doctor's non-cancelled appointments on the requested date; cancellations will no longer contribute to that count. Ties will rotate between eligible doctors within the department.

Reporting definitions

Monthly totals will count non-cancelled appointments whose appointment dates fall in the selected month. Doctor workload will show appointment count and scheduled appointment hours. Total scheduled hours will equal appointment count multiplied by the fixed duration, converted to hours; this measures booked clinical time rather than recorded attendance. Month-to-month change will compare appointment totals with the previous month. If the previous total is zero, the report will show "Not applicable" instead of dividing by zero.

5 User Scenarios

Book a department appointment

Actor Patient

Preconditions The patient is logged in. At least two Dermatology doctors are free for the requested slot.

Main flow 1. The patient asks for a Dermatology appointment and provides a date and time. 2. The system checks clinic hours and availability, compares eligible doctors' daily appointment counts, and selects the least booked doctor. 3. The patient confirms the summary; the system rechecks the slot, saves the booking, updates the doctor's schedule, and creates the calendar link.

Expected outcome One valid appointment is booked with a fair assignment, and the patient receives confirmation and a calendar link. (FR-01 to FR-05 and FR-10)

Reschedule and later cancel a visit

Actor Patient

Preconditions The patient is logged in and has a future appointment with an ENT doctor.

Main flow 1. The patient requests a new time that falls outside opening hours. 2. The chatbot explains the restriction and proposes valid slots. 3. The patient confirms an alternative; the system saves the change and updates both calendars. 4. Later, the patient confirms a cancellation, and the system releases the slot and updates the calendar event.

Expected outcome The requested changes are reflected in the patient and doctor schedules, and the cancelled slot becomes available again. (FR-03, FR-06, FR-07, and FR-10)

Review a department schedule

Actor Department Head

Preconditions The Department Head is logged in, and the department has scheduled appointments.

Main flow 1. The Department Head opens their personal weekly calendar to review booked and free slots. 2. They open the department view and compare the three doctors' appointment counts and booked hours for the same week. 3. They select a doctor to inspect that schedule; schedules outside their department remain restricted.

Expected outcome The Department Head can review personal commitments and assess workload distribution within the department. (FR-08 and FR-09)

Review monthly clinic activity

Actor Chief Medical Officer

Preconditions The Chief Medical Officer is logged in, and the system holds appointment records for the selected and previous months.

Main flow 1. The Chief Medical Officer selects a month. 2. The system calculates non-cancelled appointment totals, the percentage change from the previous month, and each doctor's workload. 3. It displays clinic-wide totals and scheduled appointment hours, using "Not applicable" when the previous month's total is zero.

Expected outcome The Chief Medical Officer can compare monthly demand and identify differences in workload across doctors and departments. (FR-11)